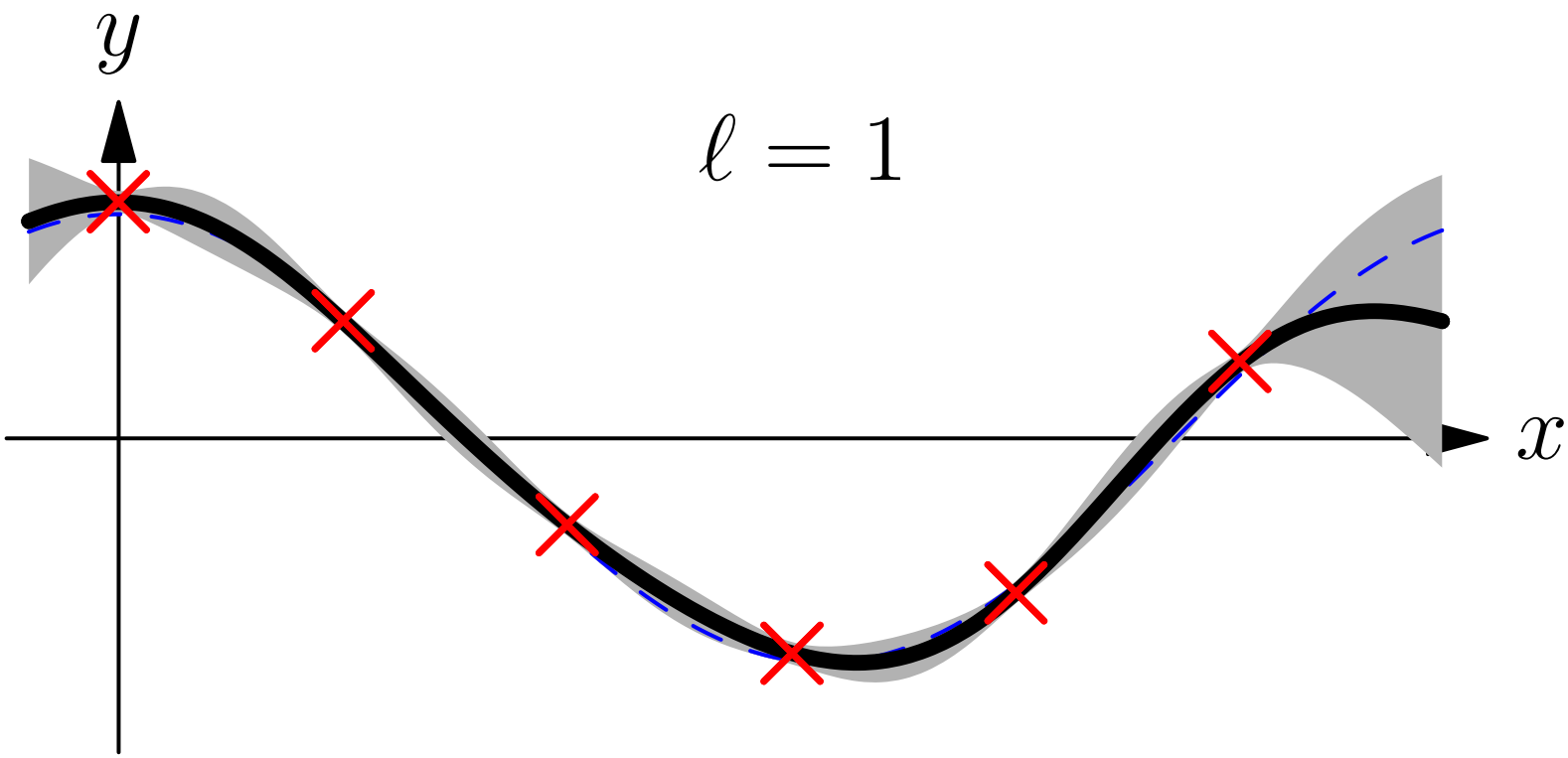
Mathematics of Machine Learning

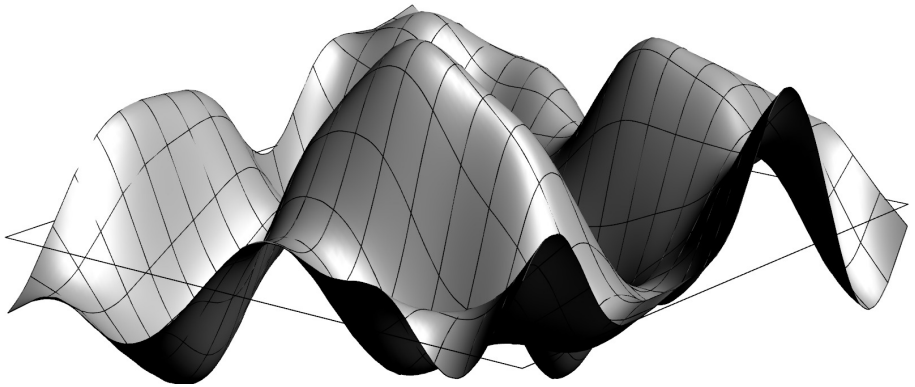
Gaussian Processes



Gaussian Processes, regression

Outline

1. **ML with Gaussians**
2. Bayesian Inference
3. Hyper-parameters
4. GPs in Practice



Machine Learning with Gaussians

- Due to the fact that we can analytically integrate over multivariate Gaussians they are heavily used in machine learning
- Examples include multivariate modelling, Kalman filters and Gaussian Processes
- Despite being analytically tractable, they are very fiddly
- This is mainly because it is easy to specify means, $\boldsymbol{\mu}$, and covariance, $\mathbf{C}$, but Gaussians involve the inverse covariance matrix

$$\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}, \mathbf{C}) = \frac{\mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\mathbf{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}}{\sqrt{\det(2\pi\mathbf{C})}}$$

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Recap: the Gaussian Toolkit

- From the last lecture, for jointly Gaussian

$$\begin{pmatrix} \boldsymbol{z}_1 \\ \boldsymbol{z}_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \right)$$

- Marginal: strike out rows and columns, $\boldsymbol{z}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$

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- Gaussian processes (GPs) are a mathematically defined ensemble of functions
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Priors on Functions

- We can think of a solution as a function $f(\boldsymbol{x})$
- We can put a prior probability distribution, $p(f)$, on a function, f, that prefers smooth functions
- We can then compute a posterior probability distribution on functions given the data, $p(f|\mathcal{D})$
- As a likelihood, $p(y_i|f(\boldsymbol{x}_i))$, we use the probability of observing y_i given the true function value is $f(\boldsymbol{x}_i)$
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Gaussian Processes

- Gaussian Processes are probability distributions over functions
- (Functions can be viewed as vectors in an infinite dimensional vector space)
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$$p(f|m,k) \propto \mathrm{e}^{-\frac{1}{2}\int (f(\boldsymbol{x})-m(\boldsymbol{x}))k^{-1}(\boldsymbol{x},\boldsymbol{y})(f(\boldsymbol{y})-m(\boldsymbol{y}))\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y}}$$

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Meaning of GP

- To understand GP's we can discretise space, $\boldsymbol{x}$, into a lattice of points $\{\boldsymbol{x}_i\}$
- Then (assuming $m(\boldsymbol{x}) = 0$)

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Covariance function

- $k(\boldsymbol{x},\boldsymbol{y})$ is a covariance function

$$\mathbb{E}\left[\left(f(\boldsymbol{x})-m(\boldsymbol{x})\right)\left(f(\boldsymbol{y})-m(\boldsymbol{y})\right)\right]=k(\boldsymbol{x},\boldsymbol{y})$$

- This is sometimes know as a **kernel**—it must be positive semi-definite
- It is a free "parameter" that the user gets to choose (although we can learn its parameters too)
- If $k(\boldsymbol{x},\boldsymbol{y})$ is a function of $\boldsymbol{x}-\boldsymbol{y}$ it is **"stationary"**
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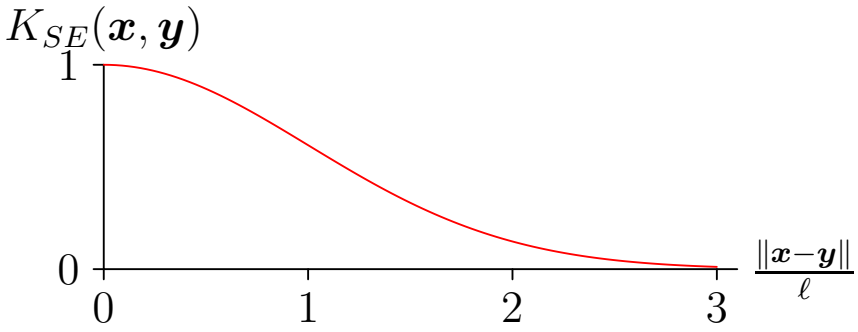
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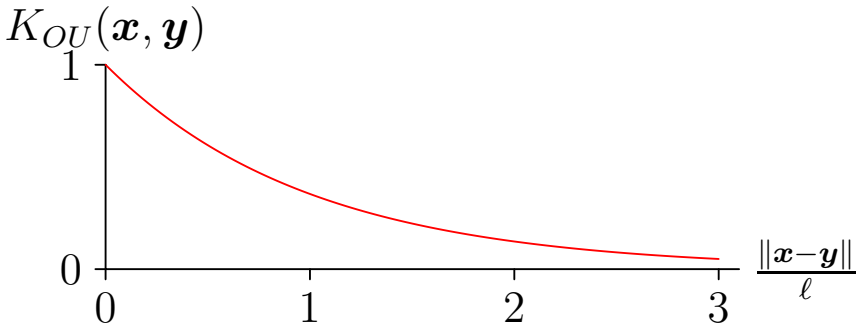
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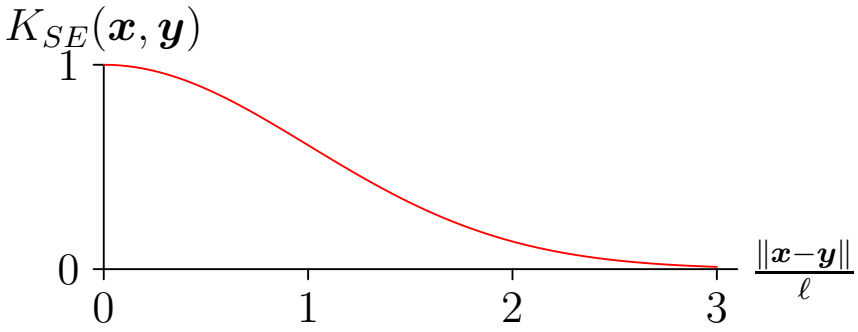
- Constant: $k_{\mathrm{C}}(\boldsymbol{x},\boldsymbol{y}) = C$
- Gaussian noise: $k_{\mathrm{GN}}(\boldsymbol{x},\boldsymbol{y}) = \sigma^2 \delta_{\boldsymbol{x},\boldsymbol{y}}$
- Squared exponential: $k_{\mathrm{SE}}(\boldsymbol{x},\boldsymbol{y}) = \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{y}\|^2}{2\ell^2}\right)$
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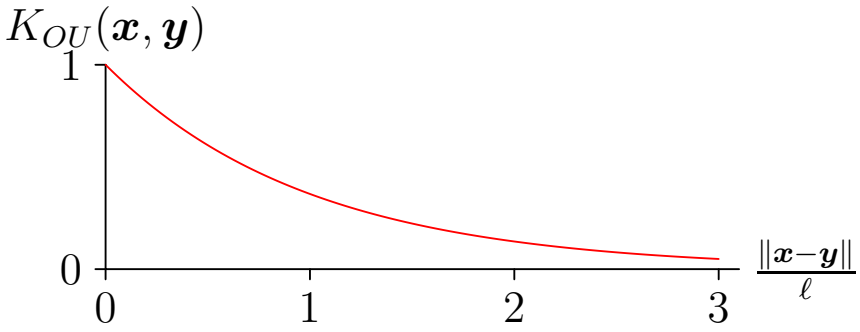




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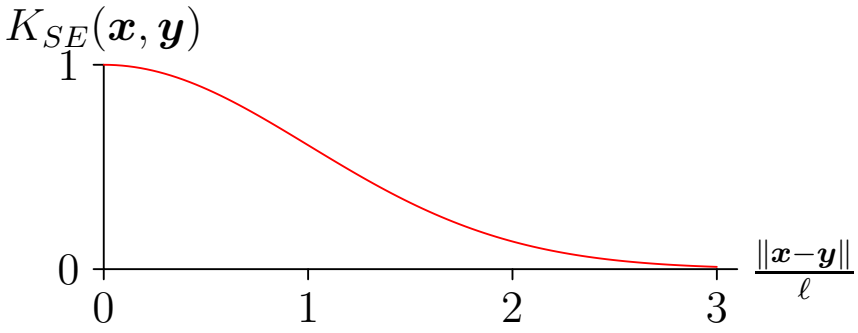
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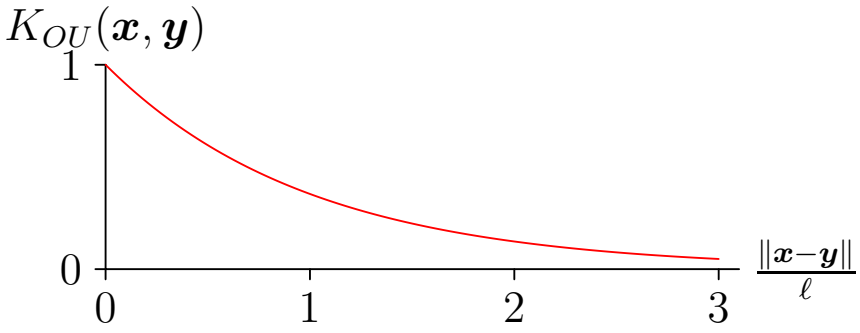




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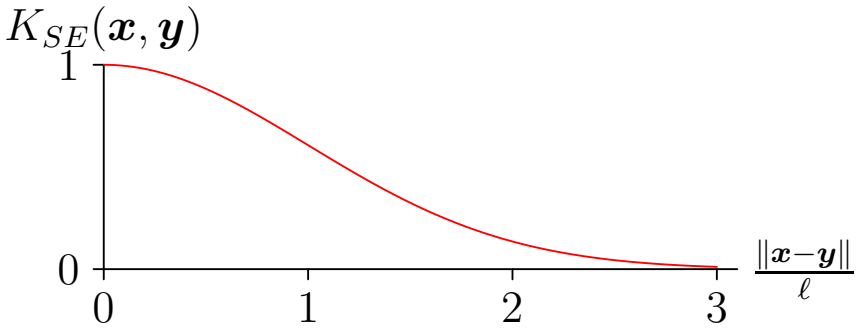
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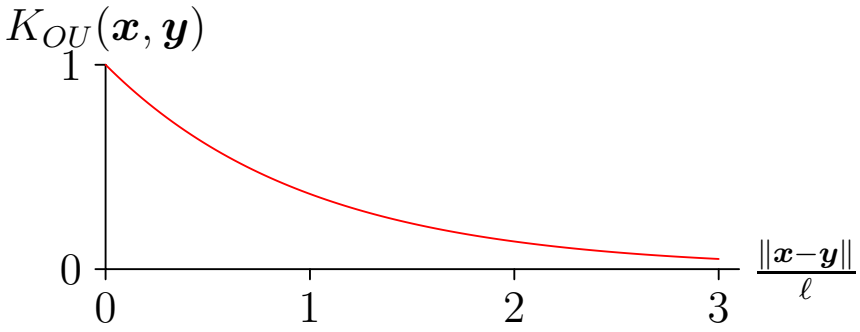




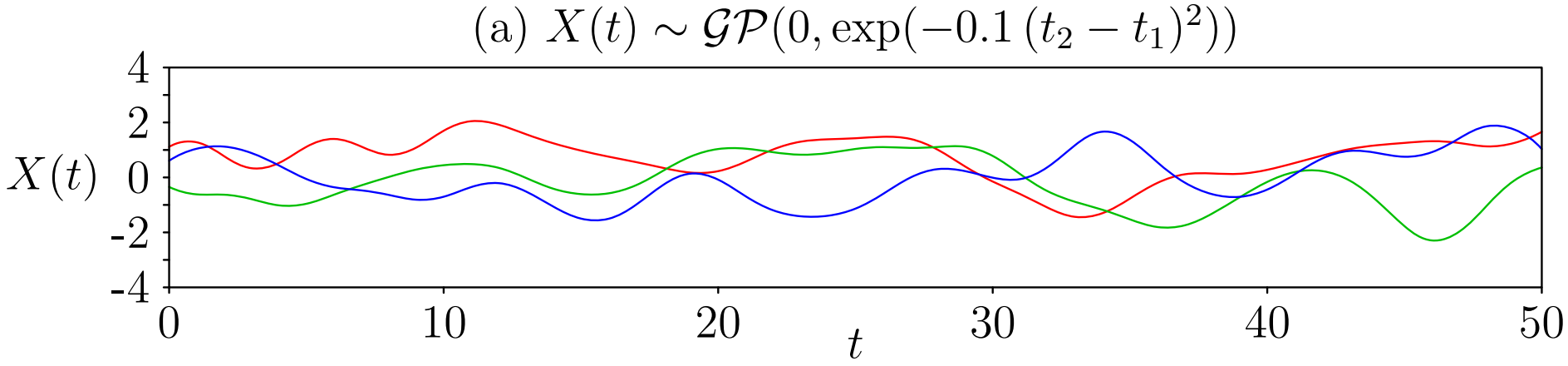
Popular Choices of GP Kernel Function

- Constant: $k_{\mathrm{C}}(\boldsymbol{x},\boldsymbol{y}) = C$
- Gaussian noise: $k_{\mathrm{GN}}(\boldsymbol{x},\boldsymbol{y}) = \sigma^2 \delta_{\boldsymbol{x},\boldsymbol{y}}$
- Squared exponential: $k_{\mathrm{SE}}(\boldsymbol{x},\boldsymbol{y}) = \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{y}\|^2}{2\ell^2}\right)$
- Ornstein–Uhlenbeck: $k_{\mathrm{OU}}(\boldsymbol{x},\boldsymbol{y}) = \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{y}\|}{\ell}\right)$

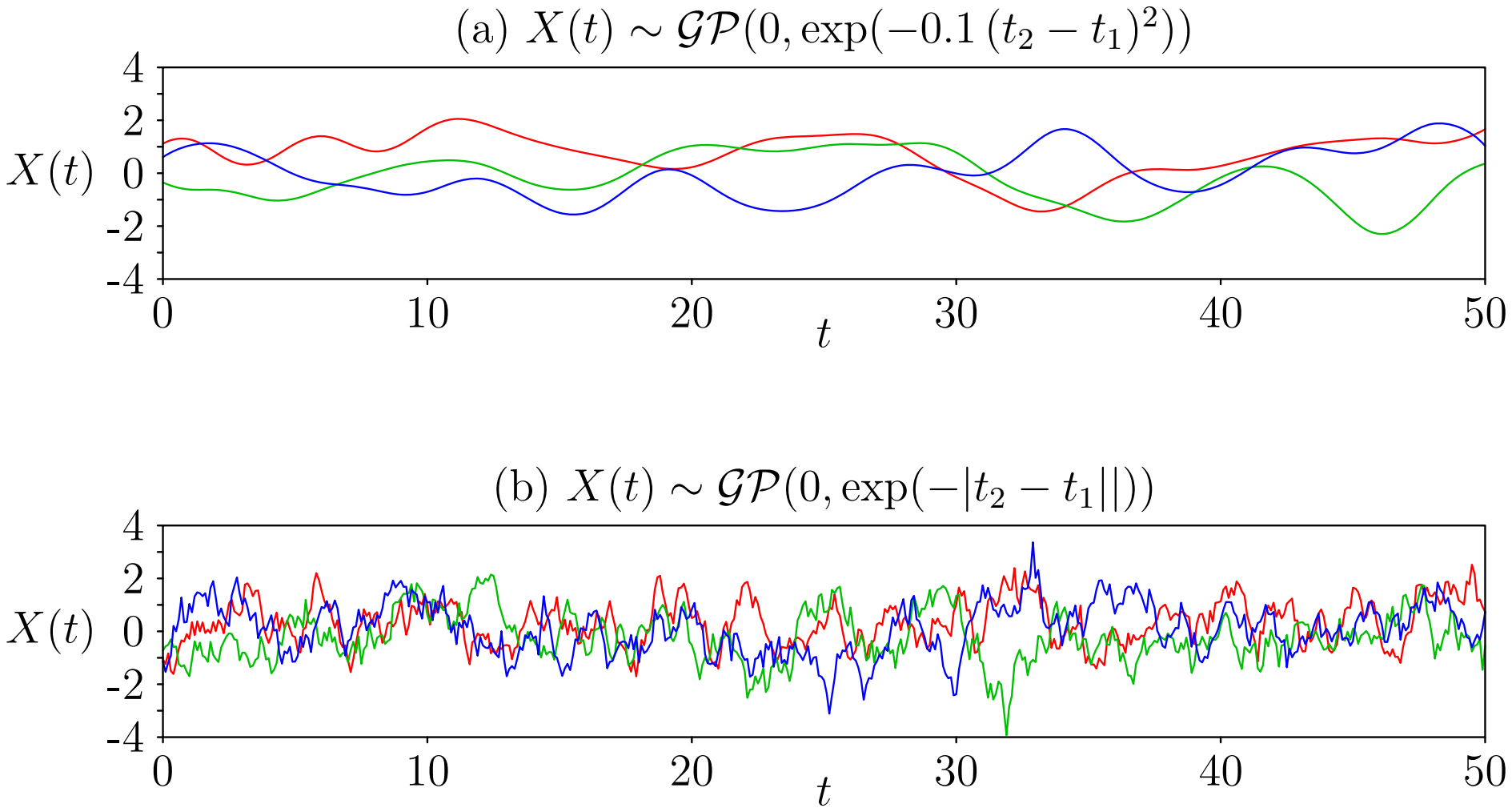




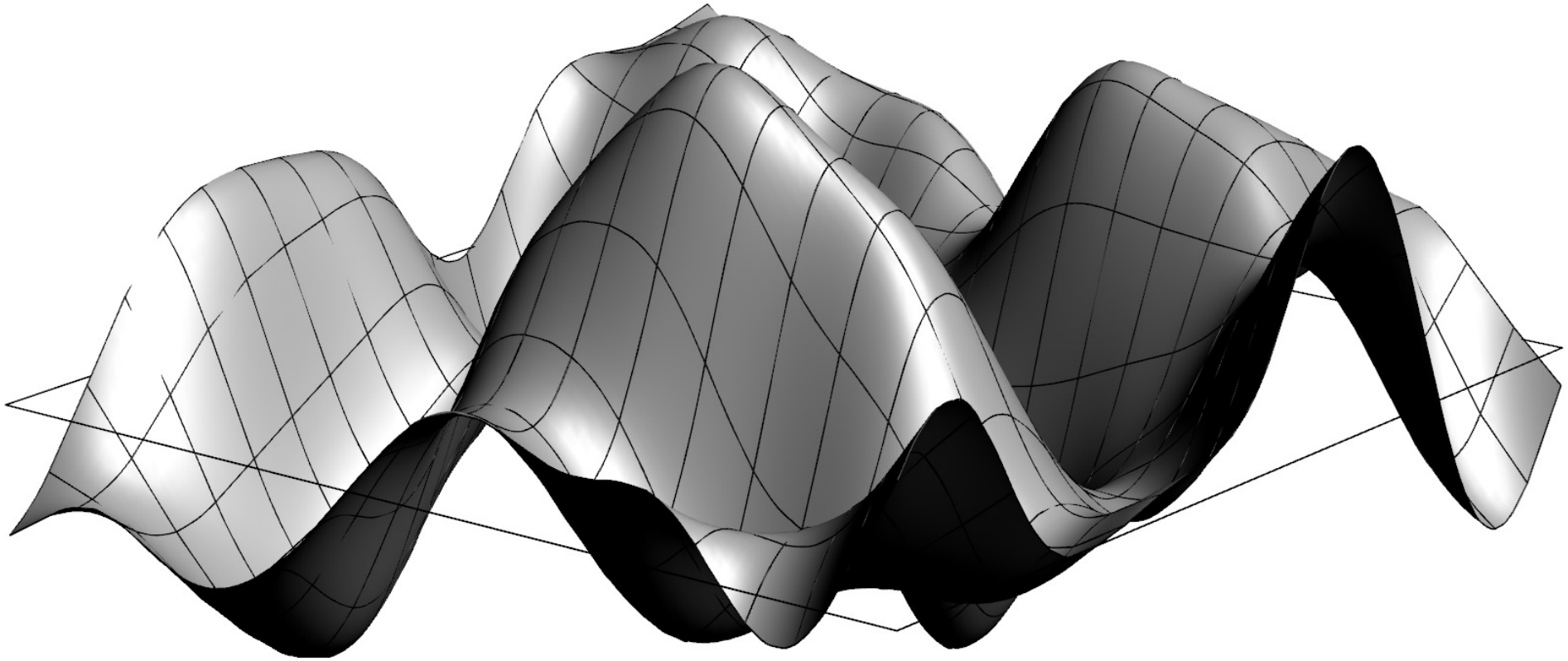
Gaussian Process Worlds



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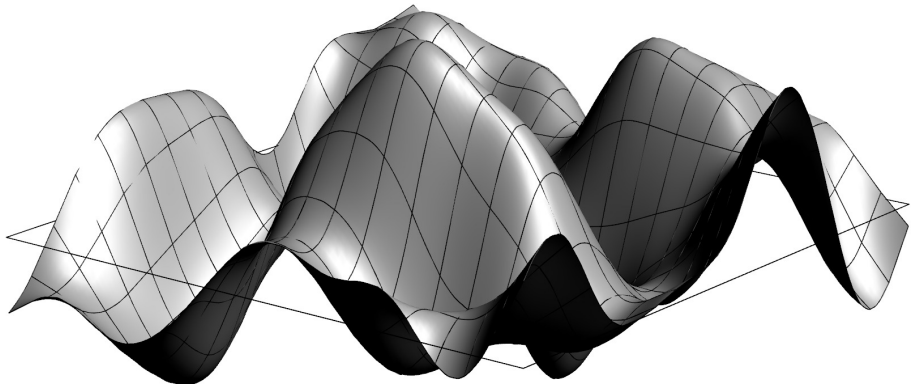


2-D Gaussian Processes



Outline

1. ML with Gaussians
2. **Bayesian Inference**
3. Hyper-parameters
4. GPs in Practice



Observed Gaussian Processes

- Given some data points $\mathcal{D} = \left((\boldsymbol{x}_i, y_i)\middle| i = 1,\ldots,m\right)$ the likelihood (assuming Gaussian error are independence of the data point) is given by

$$p(\mathcal{D}|f) = \prod_{i=1}^{m} \mathcal{N}\left(y_i\middle| f(\boldsymbol{x}_i), \sigma^2\right)$$

- Using a Gaussian Process prior we can compute a posterior using Bayes's rule

- The posterior is a Gaussian Process with a shifted mean and variance depending on the data-points

- This direct Bayesian derivation gives the answer involving the inverse matrix of the correlation function, $k^{-1}(\boldsymbol{x}, \boldsymbol{y})$

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Alternative Derivation

- Denoting the target values as a vector $\boldsymbol{y}$ with elements y_i
- Denoting the matrices of covariances between data points as $\mathbf{K}$ with elements $k(\boldsymbol{x}_i, \boldsymbol{x}_j)$
- Denoting the covariance between the data points and a particular position, $\boldsymbol{x}_*$ as $\boldsymbol{k}_*$ with elements $k(\boldsymbol{x}_i, \boldsymbol{x}_*)$
- Denoting the variance a point $\boldsymbol{x}_*$ as $k_* = k(\boldsymbol{x}_*, \boldsymbol{x}_*)$
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Conditional Probability

- To compute the posterior $p(f_*|\boldsymbol{y})$ we use

$$p(f_*|\boldsymbol{y}) = \frac{p(f_*, \boldsymbol{y})}{p(\boldsymbol{y})}$$

- where $p(\boldsymbol{y}) = \int p(f_*, \boldsymbol{y}) \mathrm{d}f_*$
- The conditional formula from the last lecture gives

$$p(f_*|\boldsymbol{y}) = \mathcal{N}\left(f_* \middle| \boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{y}, k_* - \boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{k}_*\right)$$

- Looks complicated, but numerically easy to evaluate

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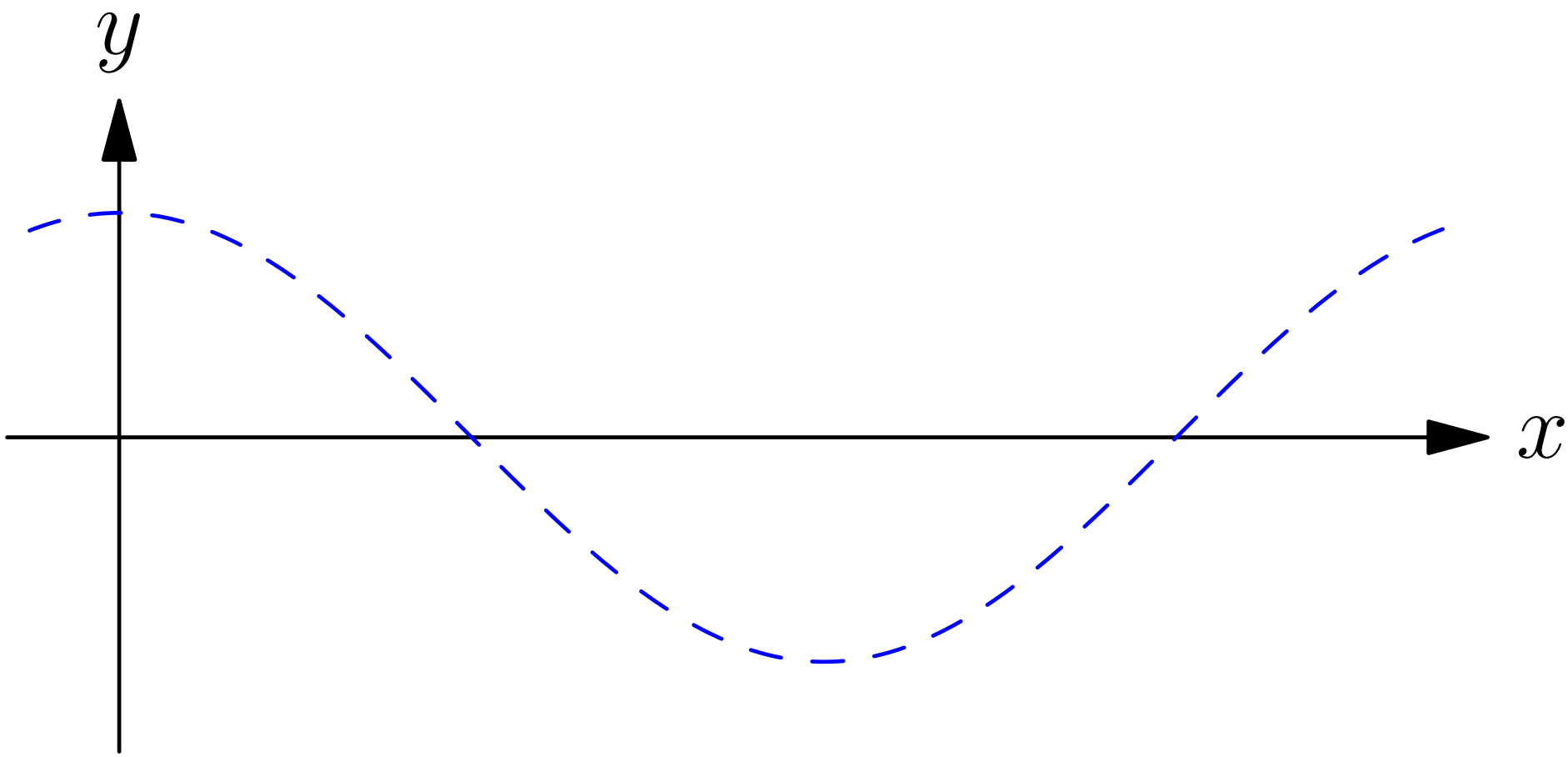
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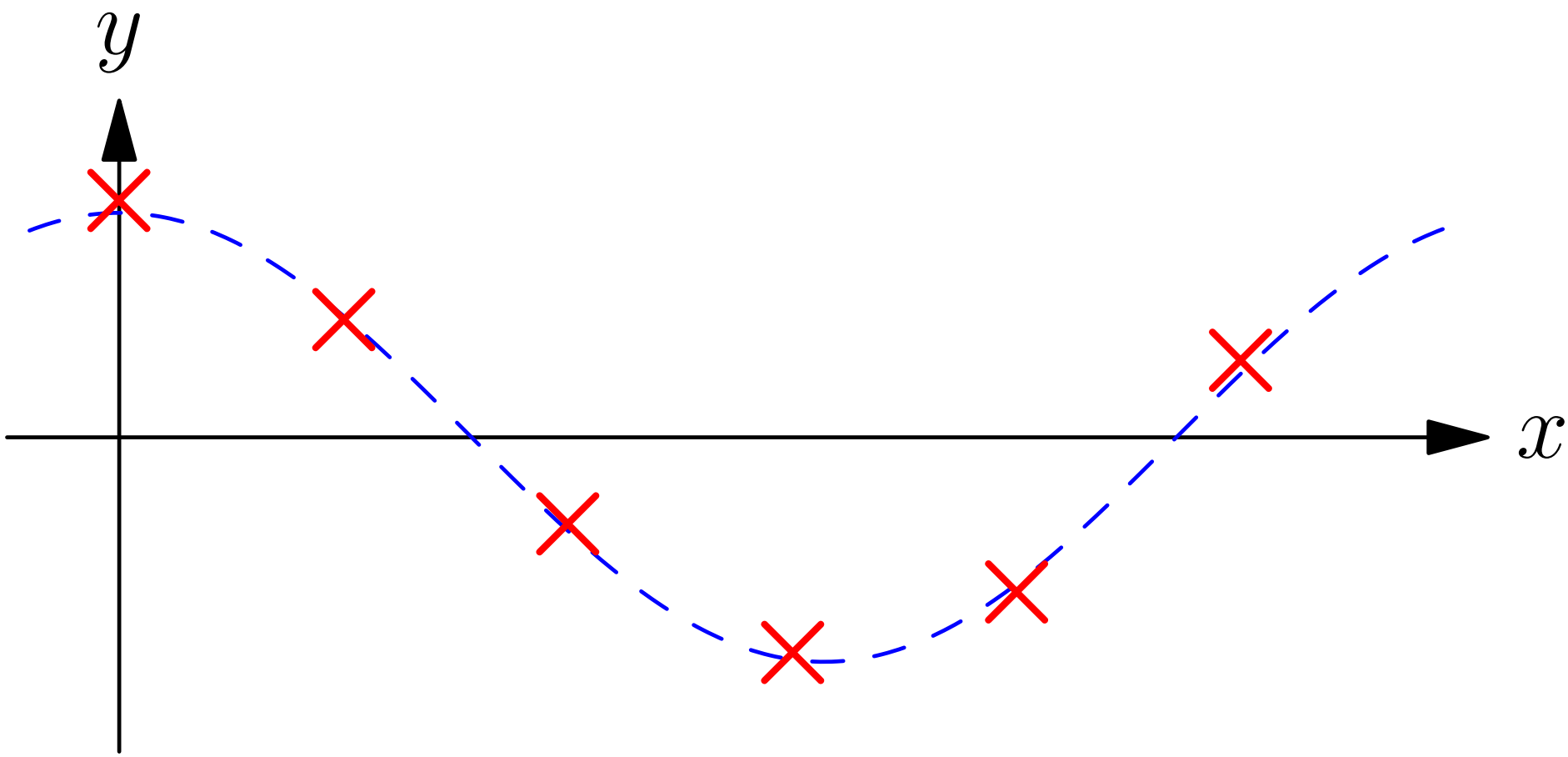
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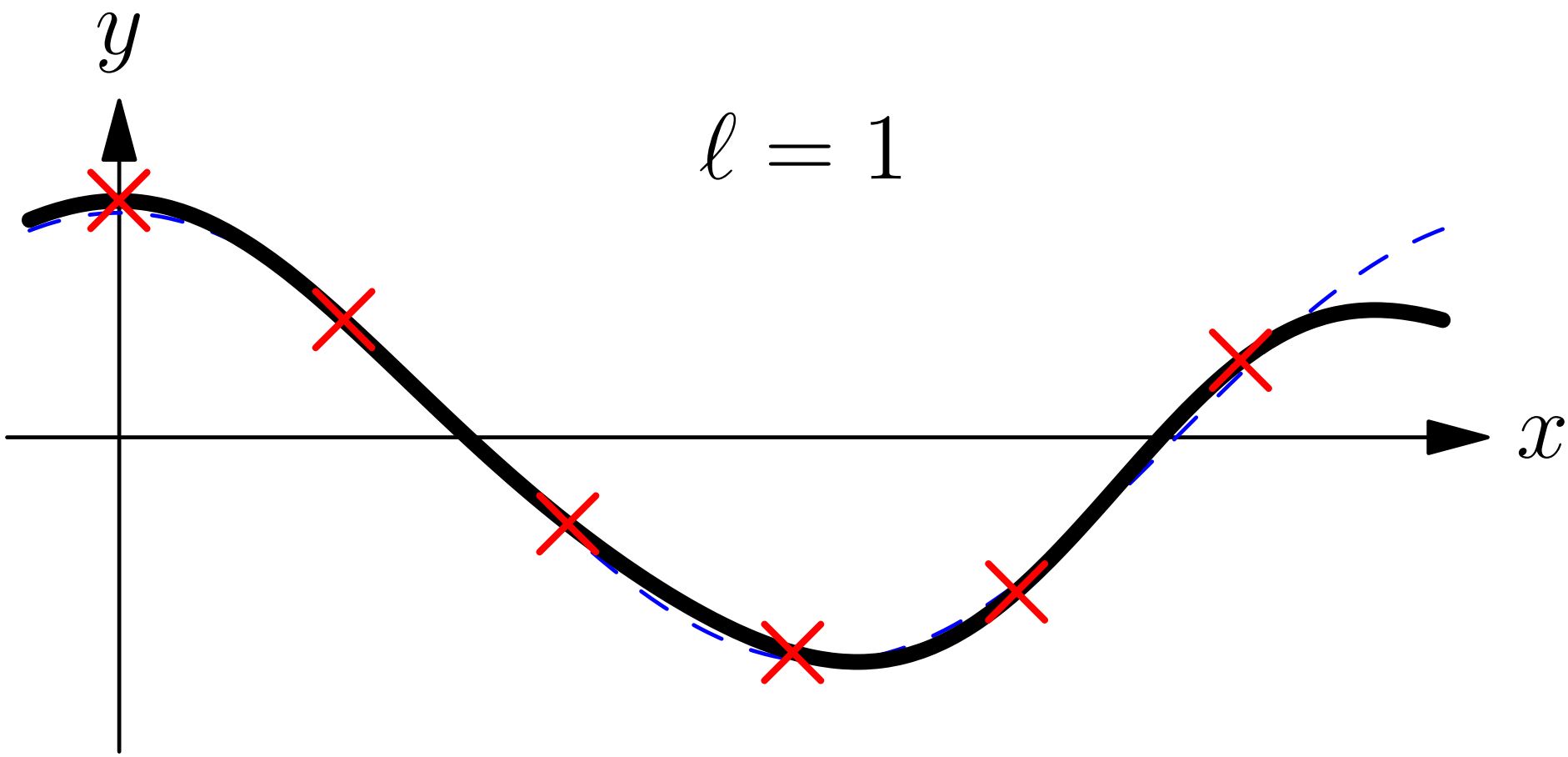
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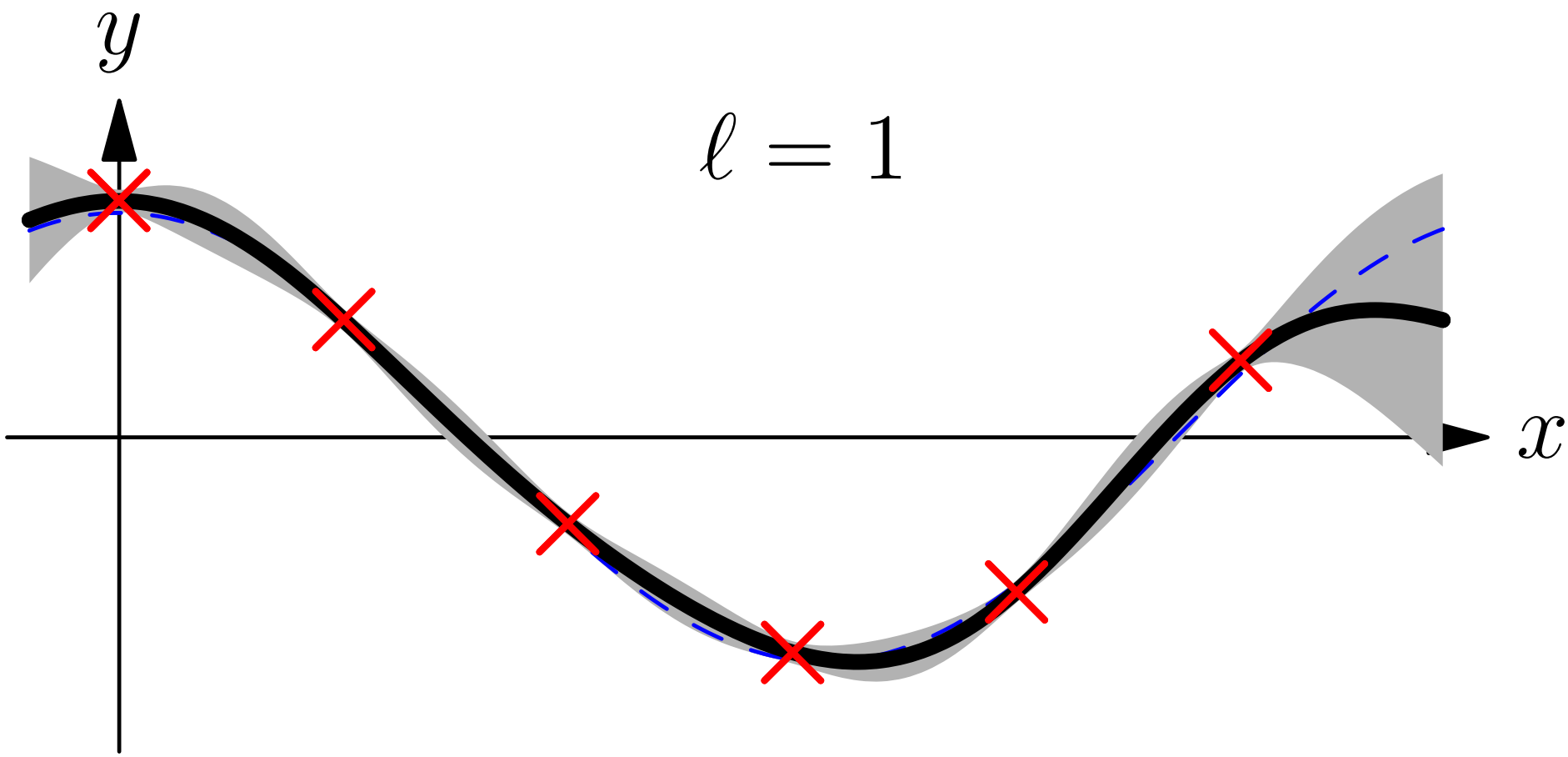
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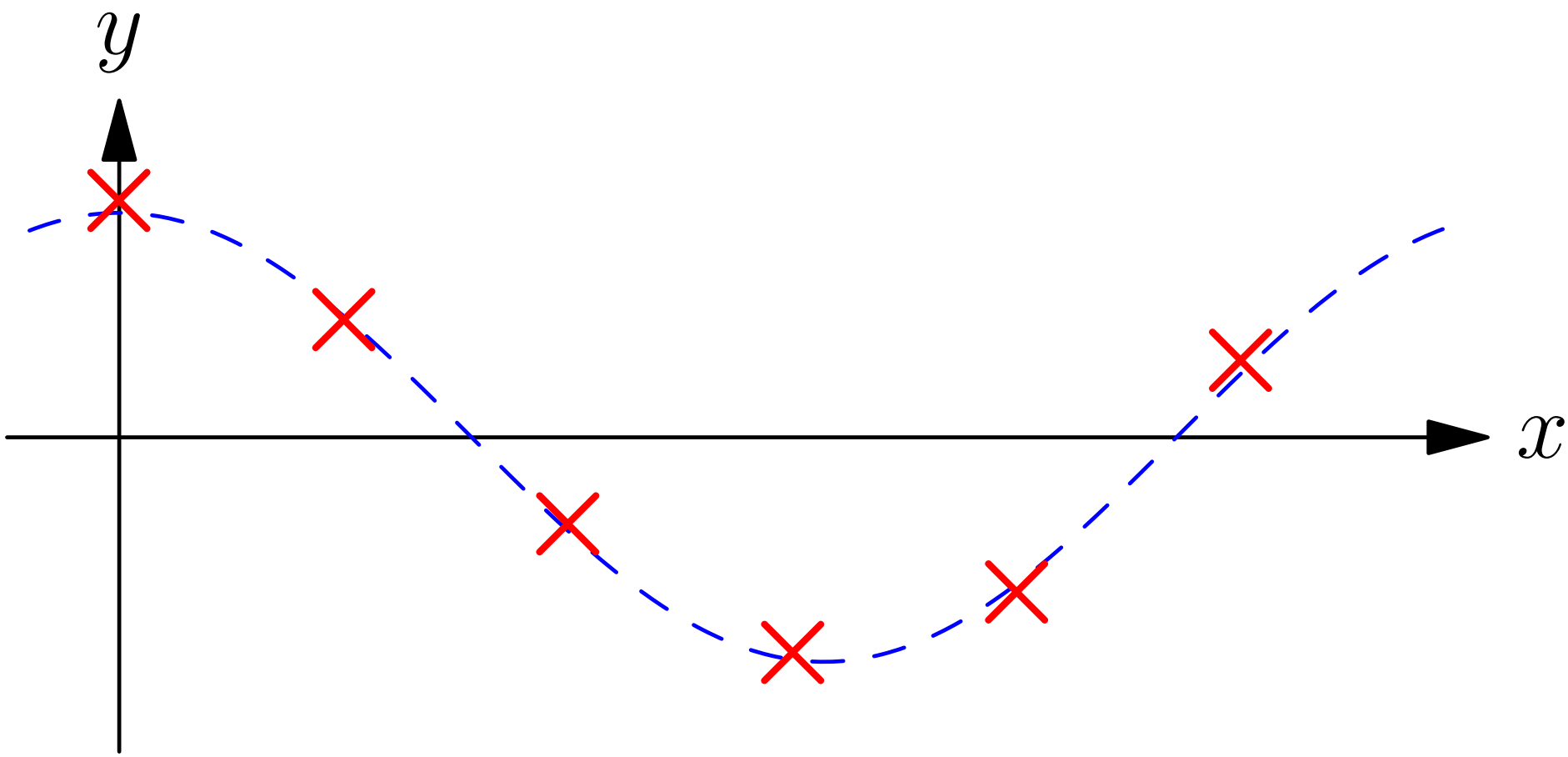
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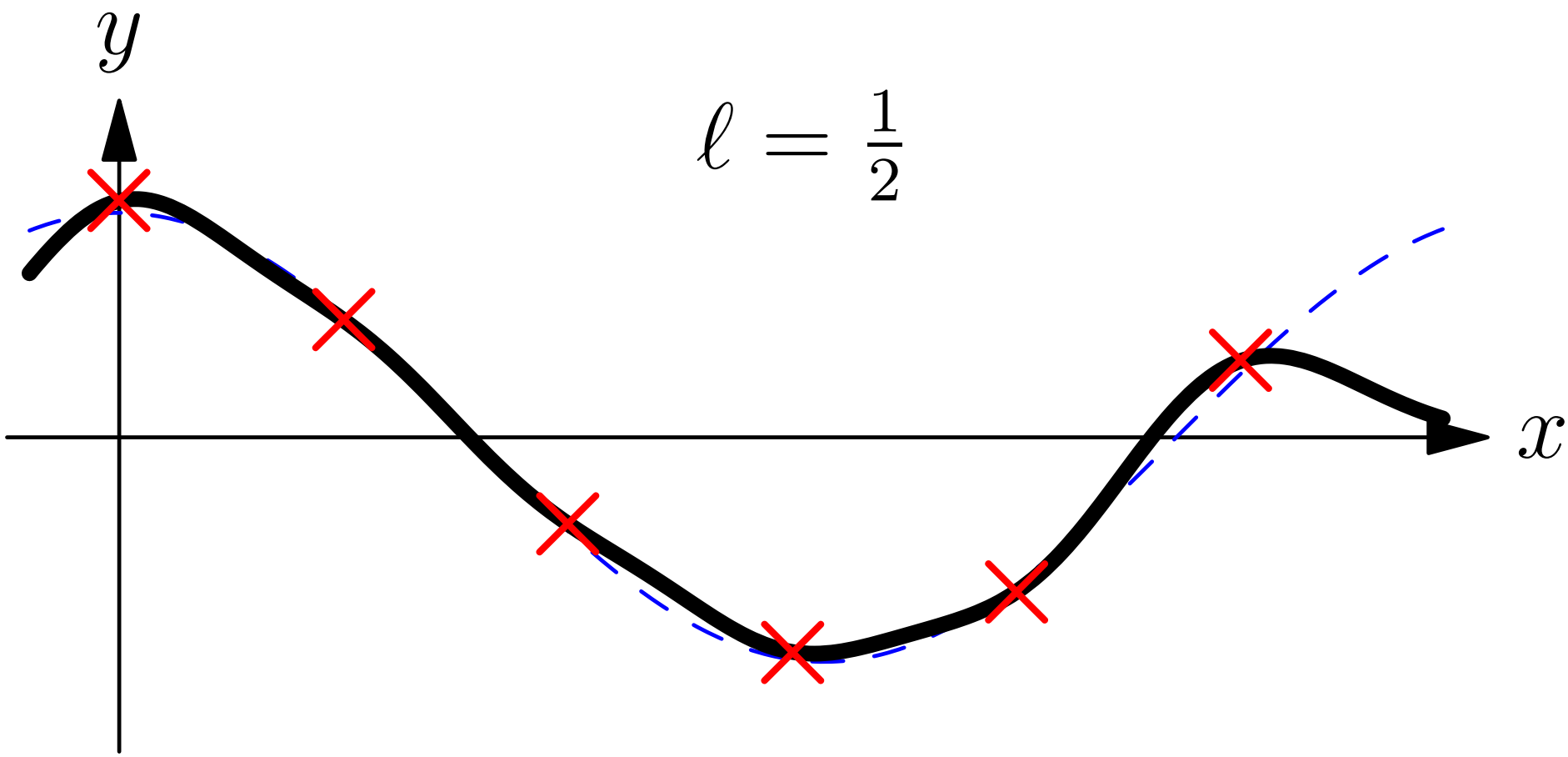
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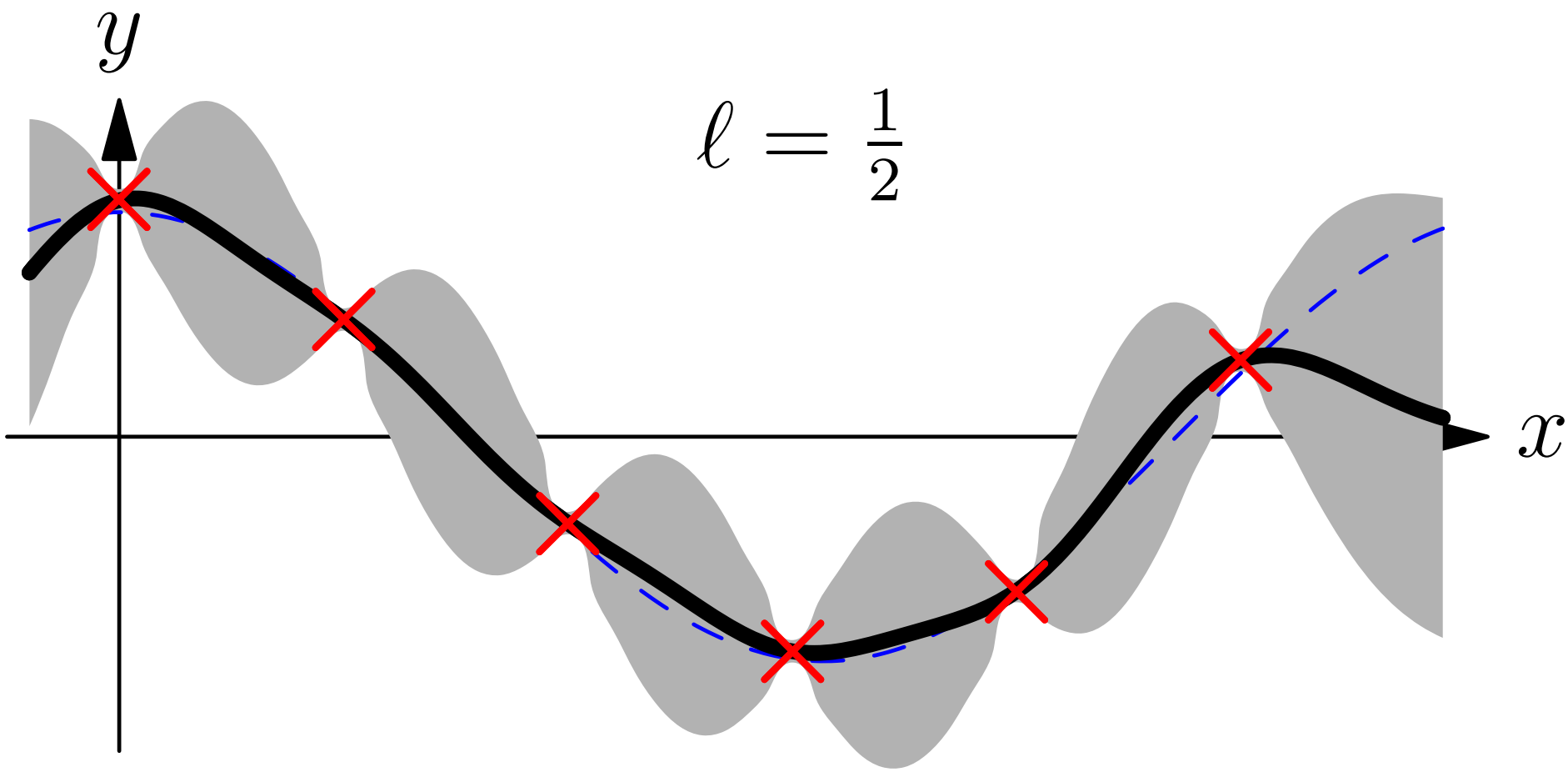
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Multi-dimensional Regression

- I've shown a 1-D regression example because it is easy to visualise
- This might be used with a time series
- The much more typical situation in machine learning is for $\boldsymbol{x}$ to have many features so we are doing multi-dimensional regression
- Gaussian process inference were first used in spatial problems where it was known as **krigging**
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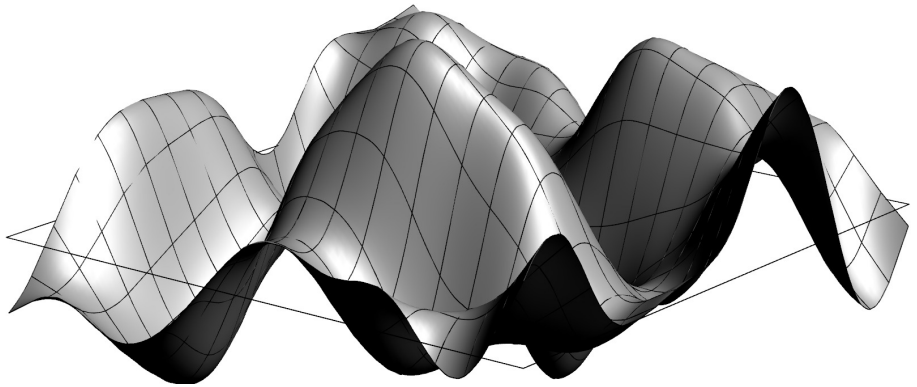
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Outline

1. ML with Gaussians
2. Bayesian Inference
3. **Hyper-parameters**
4. GPs in Practice



Choosing the Correct Covariance Function

- Choosing the correct covariance function is critical
- Most covariance functions include a continuous **hyper-parameter** (e.g. the correlation length ℓ) that we have to choose correctly
- This is typical of many Bayesian problems where we have some set of hyper-parameters, $\boldsymbol{\phi}$, describing the model
- These are different to the normal parameters we learn (e.g. weights $\boldsymbol{w}$ or in GP the functions $f(\boldsymbol{x})$)
- In Bayesian inference we learn the posterior for these normal parameters

$$p(f|\mathcal{D},\boldsymbol{\phi}) = \frac{p(\mathcal{D}|f,\boldsymbol{\phi})p(f|\boldsymbol{\phi})}{p(\mathcal{D}|\boldsymbol{\phi})}$$

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Evidence Framework

- The normalisation factor, $p(\mathcal{D}|\boldsymbol{\phi})$ is known as the **marginal likelihood** or **evidence**

$$p(\mathcal{D}|\boldsymbol{\phi}) = \int p(\mathcal{D}|f,\boldsymbol{\phi})p(f|\boldsymbol{\phi})\mathrm{d}f$$

- We can perform a Bayesian calculation at a second level by putting a prior on $\boldsymbol{\phi}$

$$p(\boldsymbol{\phi}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\phi})p(\boldsymbol{\phi})}{p(\mathcal{D})}$$

- From this we can now marginalise out the hyper-parameters

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  usually can't be computed analytically and we have to use Monte Carlo methods (see later lecture)

- An alternative is to use the most likely hyper-parameter
- We can find this by using gradient search of $p(\mathcal{D}|\boldsymbol{\phi})$
- This is sometimes referred to as ML-II
- Normally even this can be difficult, but for GP its not too difficult

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Evidence for GP

- For GP the (log)-evidence can be computed in closed form

$$\log(p(\mathcal{D}|\boldsymbol{\phi})) = -\frac{1}{2}\boldsymbol{y}^{\mathsf{T}}(\mathbf{K}+\sigma^2\mathbf{I})^{-1}\boldsymbol{y} - \frac{1}{2}\log\left(|\mathbf{K}+\sigma^2\mathbf{I}|\right) - \frac{m}{2}\log(2\pi)$$

  - ⋆ First term measures goodness of fit
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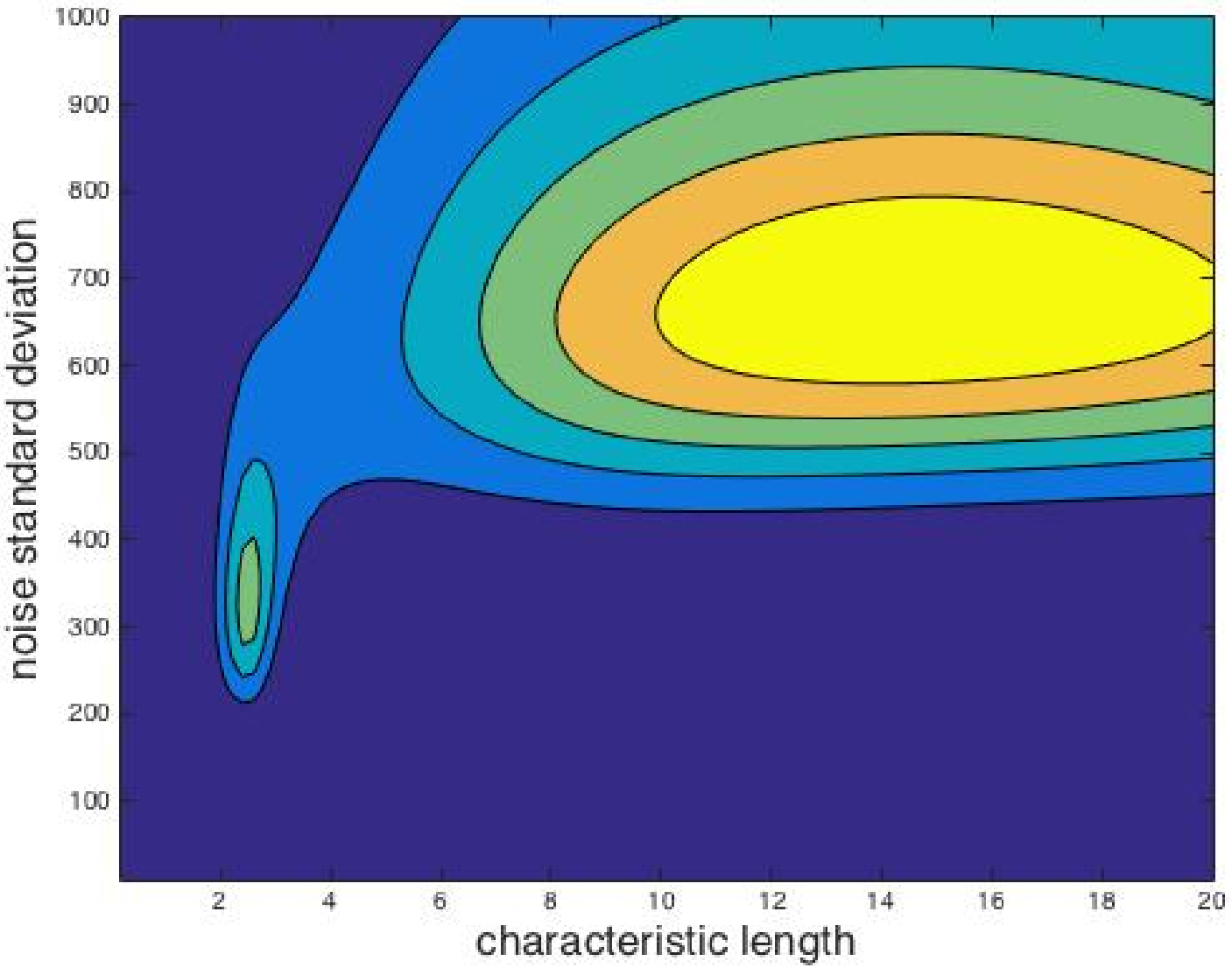
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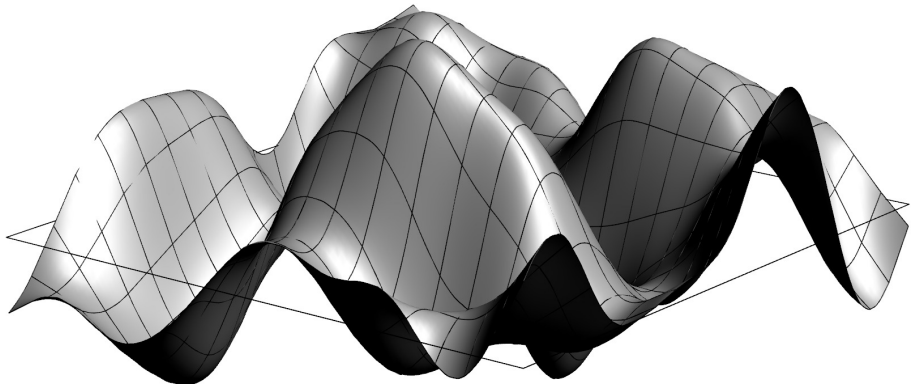
- Could overfit!

Example (slightly pathological)

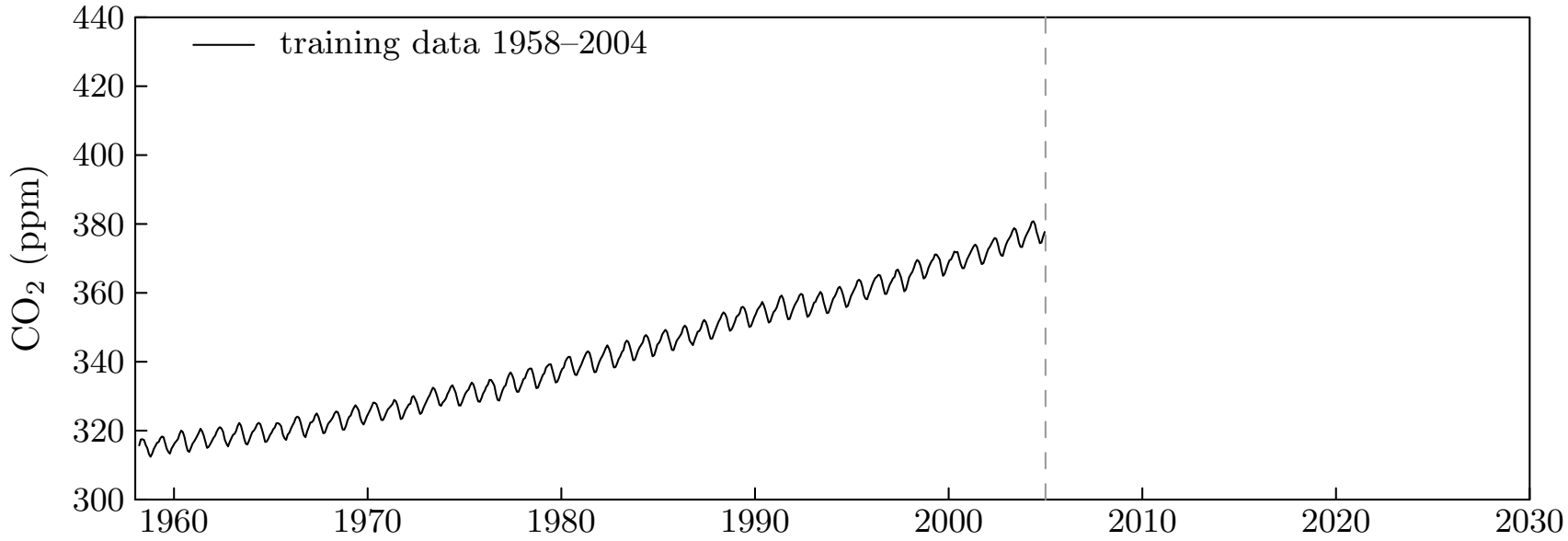


Outline

1. ML with Gaussians
2. Bayesian Inference
3. Hyper-parameters
4. **GPs in Practice**

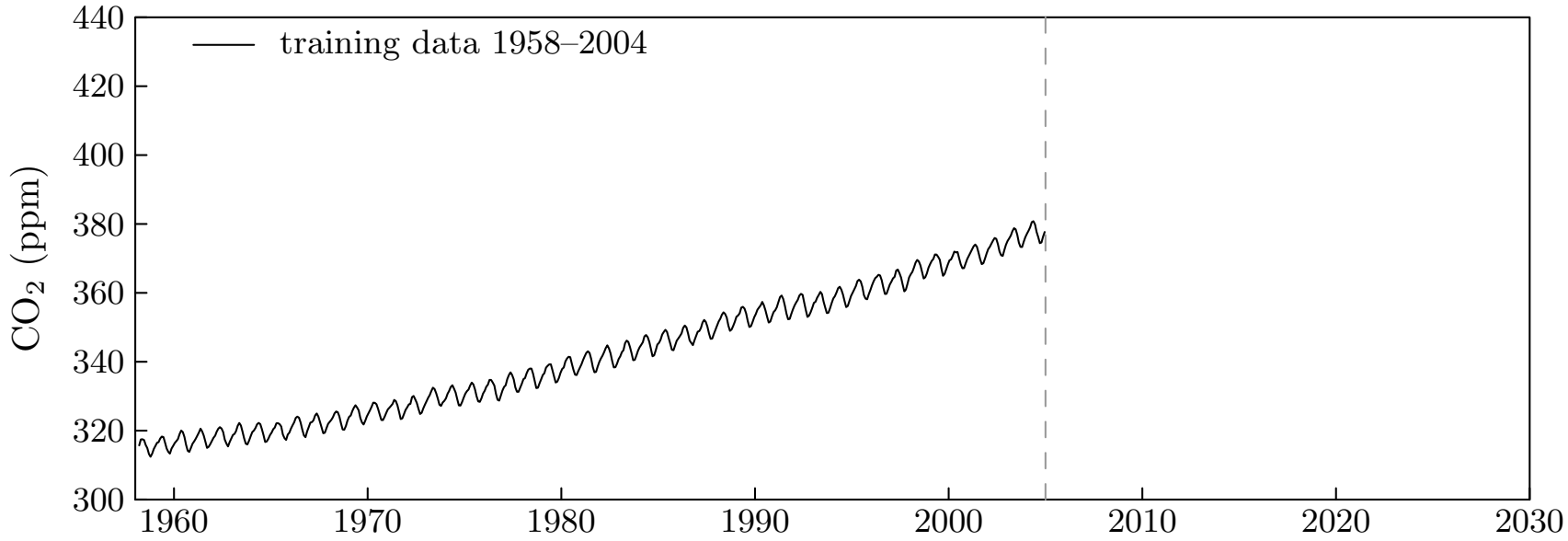


Example: CO_2 at Mauna Loa



- Monthly mean CO_2 since 1958 (data: NOAA GML and Scripps)
- Train on 1958–2004 (562 months), forecast to 2030, and check against what was actually measured

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Building the Kernel

- Look at the data and add one term for each feature you see ($r = t - t'$ in years)
  - ⋆ a rising trend: a quadratic with uncertain coefficients, $c_0^2 + c_1^2 t t' + c_2^2 t^2 t'^2$, plus a smooth $s_1^2 k_{\mathrm{SE}}(r; \ell_1)$
  - ⋆ a yearly cycle that slowly changes shape: $s_2^2 \mathrm{e}^{-r^2/(2\ell_2^2)} \mathrm{e}^{-2\sin^2(\pi r)/\ell_3^2}$
  - ⋆ short wiggles $s_3^2 k_{\mathrm{SE}}(r; \ell_4)$, and measurement noise $\sigma^2 \delta_{t,t'}$
- Sums and products of kernels are kernels (kernel trick lecture)

- The period is fixed at one year: use what you know

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Fitting the Hyperparameters

- Eleven hyperparameters, found by maximising the log evidence with a gradient method
- Work with their logarithms (they must be positive), start from sensible values (ppm, years), and use several restarts
- The fitted values make sense: the cycle changes shape over about a century ($\ell_2 \approx 107$ years), the wiggles last about seven weeks ($\ell_4 \approx 0.14$ years), the noise is $\sigma \approx 0.19$ ppm
- The evidence also compares models: log evidence -126 with the quadratic trend against -151 without it
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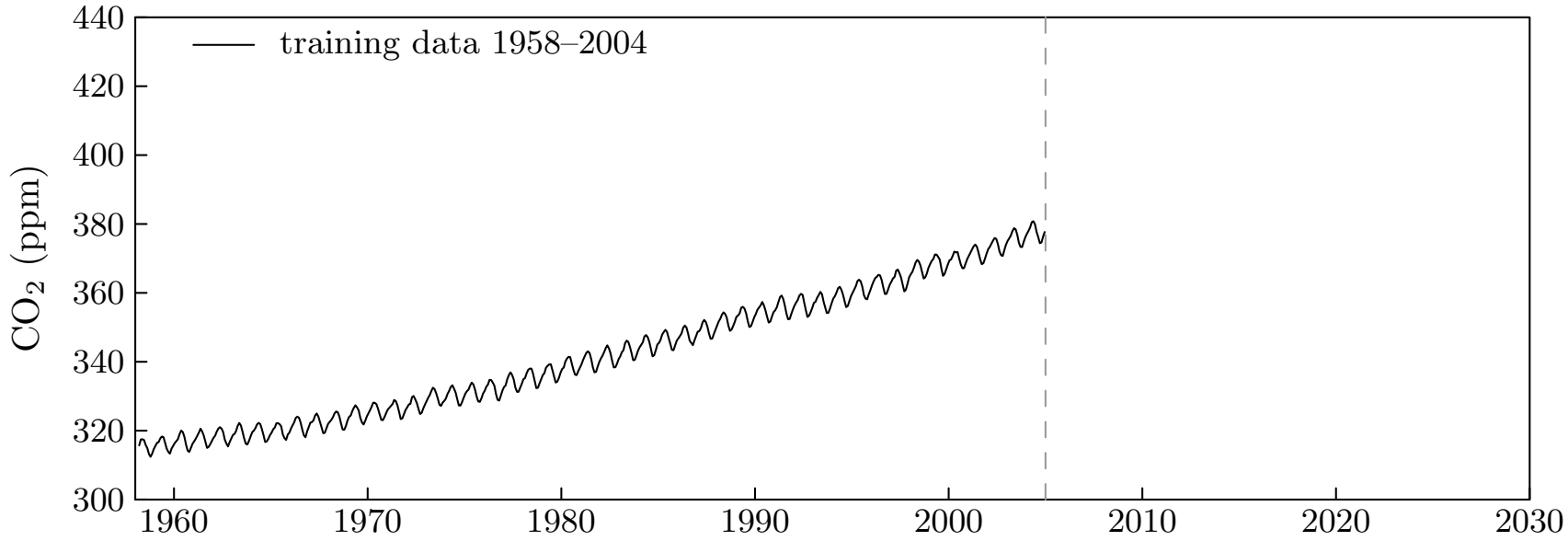
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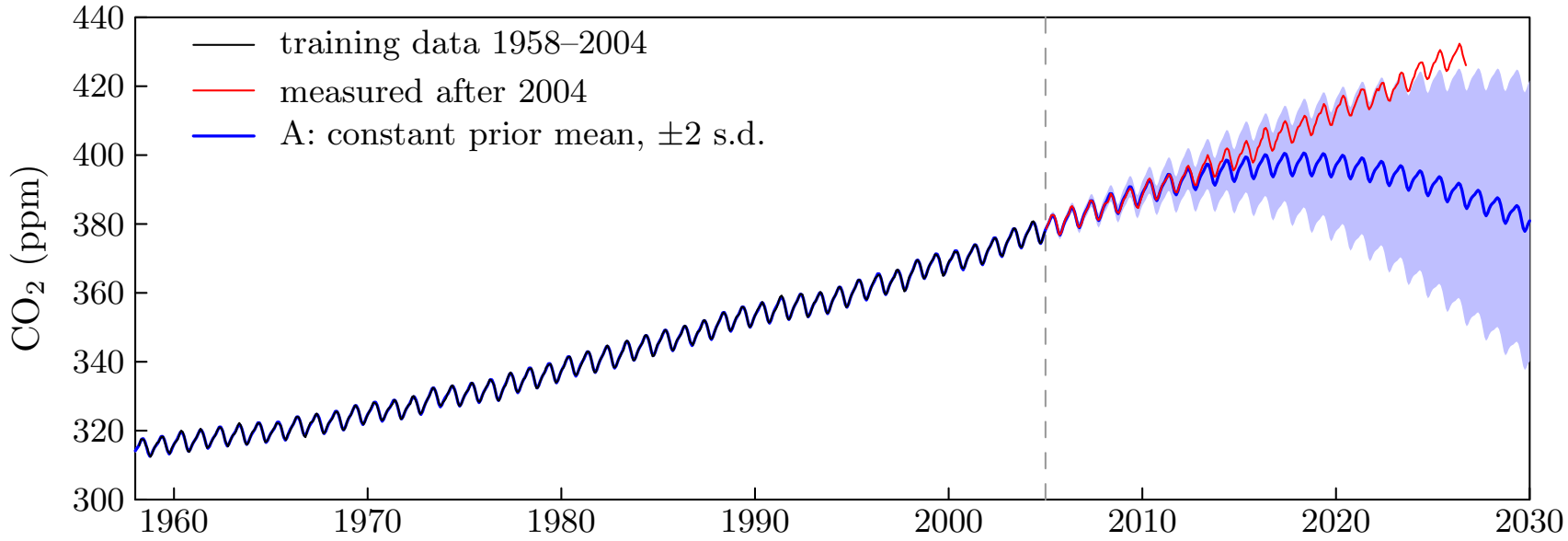
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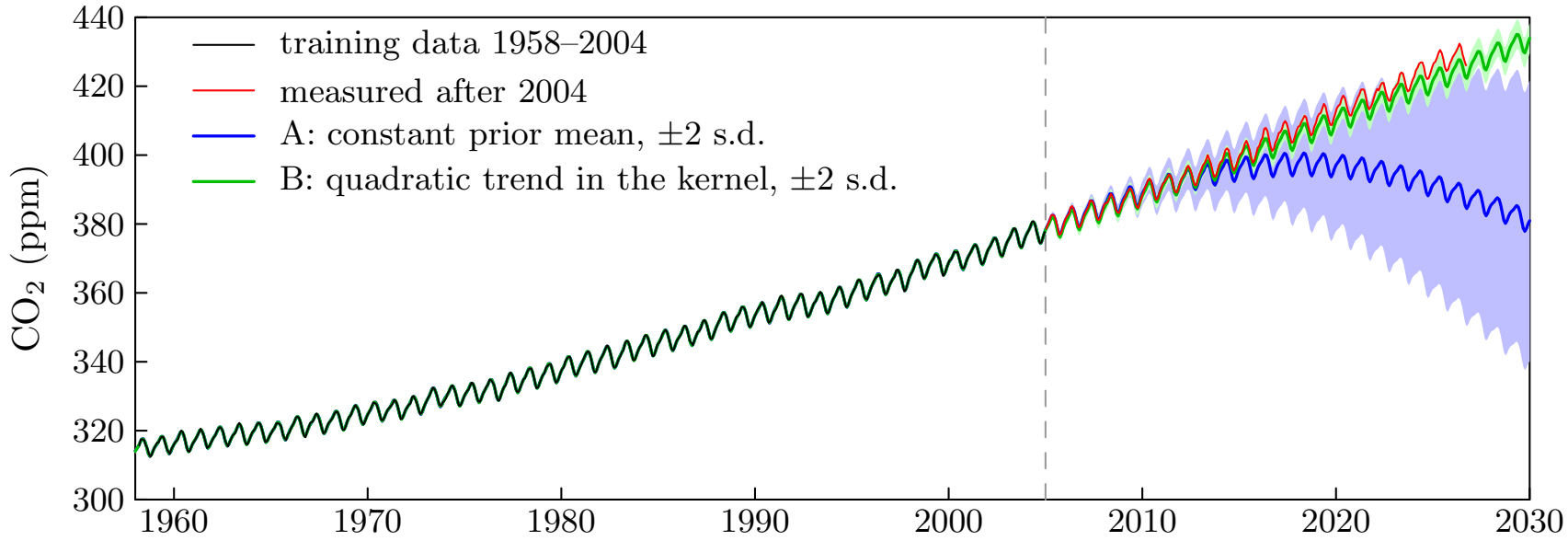
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Making GPs Work

- Build the kernel from what you know: smooth (SE), rough (OU), repeating (periodic), trends (polynomial); combine with sums and products
- Put trends into the model: away from the data a GP falls back to its prior mean
- Standardise inputs and outputs, fit the hyperparameters by the evidence with several restarts, and check they make sense
- Test on held-out data, especially when extrapolating
- Exact GPs cost $O(m^3)$: fine up to about 10^4 points, beyond that use sparse approximations
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